Stroke-Level Online Handwriting Synthesis

A Vector-Output Generative Sequence Architecture for Composable Handwritten Documents

CORE GENERATIVE PIPELINE

Arbitrary Text String → Soft-Window Attention → Autoregressive Temporal Strokes → C^1 Bézier SVG

70/70 Unit Tests Green

-2.155 NLL Loss

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PROGRAMME

B.Tech CSE

Specialization in AI
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MENTORS & REVIEWER

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INSTITUTION

GLA University

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Session 2026–2027

Why Online Vector Synthesis?

Overcoming the fundamental flaws of static fonts and raster 2D diffusion models.

1. Static Digital Fonts

`.TTF / .OTF`

Every instance of a letter renders identical fixed outlines.

- **Repetition Flaw:** Exact duplicate glyphs across a document; immediately detected as artificial.
- **Zero Ligatures:** Cannot model organic connecting strokes between cursive letters.
- **Rigid Baselines:** Lacks natural human drift, slant variation, and pressure cadence.

2. 2D Raster Models

`Diffusion / GANs`

Synthesizes handwriting directly as static RGB pixel bitmaps.

- **Resolution Barrier:** Fixed 256×256 or 512×512 grids; severe blur when zoomed in.
- **Heavy Latency:** Multi-step denoising requires massive GPU compute per sentence.
- **No Vector Control:** Output is a baked raster; strokes cannot be parsed or edited.

3. Online Stroke Synthesis

`hws (Our Project)`

Generates temporal pen coordinates `(Δx, Δy, Δt, p)` sequentially.

- **Infinite Scalability:** Renders to scalable vector graphics (`SVG` `<path>`); razor sharp at any zoom.
- **Sub-3s Inference:** Lightweight autoregressive sequence decoding on edge GPUs.
- **Authentic Physics:** Models cursive ligatures, airborne transitions, and notebook ruling.

Key Insight: Handwriting is fundamentally a *continuous temporal trajectory* of a physical stylus tip on paper, not a 2D image. Modeling the time-series trajectory yields mathematical elegance, infinite zoom, and compact storage.

`Δs = 2.0 Spatial Arc-Length`

Algorithmic Innovations & Data Pipeline

Solving Graves 2013 variance collapse, sequence bloat, and ragged dataset storage.

12,178	3,733,404	92.1% / 7.9%	1.84×
IAM-ONDB LINES	STROKE POINTS	MOVES VS. AIRBORNE JUMPS	SEQUENCE COMPRESSION

Decision 1 (D1): Dual-Mode Tokenization

Stable Cross-Entropy

Continuous Mixture Density Networks (MDNs) suffer from numerical instability where variance collapses ($\sigma \to 0$, log-loss $\to \infty$). We partition the velocity space into two distinct regimes:

```
Airborne Jumps (asinh Binning):
k = sgn(Δx) · ⌊ asinh(|Δx| / s) · N ⌋
```

- **Moves (Pen Down):** High-density linear spatial bins capturing delicate cursive curves.
- **Jumps (Pen Up):** Heavy-tailed inverse hyperbolic sine (asinh) bins covering wide transitions.
- **Stability:** Replaces fragile Gaussian log-likelihood with bounded Cross-Entropy loss.

Decision 2 (D2): Ragged HDF5 & Resampling

Storage & Throughput

Raw tablet digitizer streams sample points at erratic temporal intervals, leading to dense clusters during slow writing and sparse gaps during fast strokes.

- **Spatial Arc-Length Resampling:** Enforces uniform Euclidean spacing `ds = 2.0` along the stroke trajectory.
- **1.84× Sequence Reduction:** Compresses average line length from 620 down to 342 points without losing fidelity.
- **Ragged 1D HDF5 Arrays:** Stores variable-length samples contiguously in 1D memory `(N_total, 4)` with index table `(M, 2)`.
- **Zero Waste:** Eliminates 40% wasted disk space compared to fixed zero-padded matrices.

Dual-Track Neural Sequence Modeling

Reproducing the canonical Graves 2013 recurrent baseline alongside modern Transformer attention.

1. **Text Conditioning** — One-Hot / Tokenizer
2. **Monotonic Attention** — Soft-Window (κ_t)
3. **Recurrent / Trans State** — Peephole LSTM / Causal
4. **Output Heads** — 20-Comp MDN / Discrete
5. **SVG Render Engine** — Catmull-Rom Splines

Track A: Canonical Graves 2013 RNN

Baseline Reproduction

3-layer recurrent neural network with cell-to-gate peephole connections and monotonic Gaussian soft-window attention:

$$\kappa_t^k = \kappa_{t-1}^k + \exp(\alpha_t^k), \quad w_t(u) = \sum \alpha_t^k \exp(-\beta_t^k(\kappa_t^k - u)^2)$$

- **Monotonic Guarantee:** Window center κ only advances forward; strictly prevents reading backward.
- **20-Component Bivariate GMM:** Predicts (μ_x, μ_y, σ_x, σ_y, ρ, π) plus Bernoulli pen lift.
- **Reproduction Status:** Reached **-2.155 NLL validation loss** matching published SOTA.

Track B: Autoregressive Transformer

Custom Innovation

Bidirectional text encoder coupled with a causal autoregressive Transformer decoder utilizing multi-head cross-attention:

- **Causal Self-Attention:** Attends to previous stroke trajectory tokens without recurrent bottleneck.
- **Cross-Attention:** Dynamically aligns stroke coordinates to text characters across all positions.
- **Style Conditioning (e_{style}):** Incorporates learned writer embeddings to synthesize varied personal handwriting aesthetics.
- **Training Milestone:** 20 full epochs trained (`checkpoints/best_model.pt`).

Visual Quality & Calligraphic Stability

Eliminating Brownian hand tremors, robotic polygon edges, and character skipping.

1. Temperature Scaling (T = 0.35)

Tremor Elimination

Unconditioned Gaussian sampling exhibits severe Brownian high-frequency noise resembling severe muscle tremors.

$$\sigma_{scaled} = \sigma \cdot T^{1.2} \quad (T = 0.35)$$

Shrinking mixture covariance concentrates sampling around the modal skeleton of letterforms while preserving organic cursive curvature.

2. C^1 Catmull-Rom Bézier Splines

Fountain Pen Flow

Standard polyline renderers draw rigid straight line segments (`L` commands) that appear segmented and polygonal.

- Interpolates discrete coordinates into smooth cubic Bézier curves (`C` commands in SVG).
- Guarantees continuous first-order tangents (C^1) across every waypoint.
- Transforms raw discrete coordinates into fluid ink calligraphy.

3. Savitzky-Golay Denoising

window=7, poly=3

Digital low-pass moving-average filters destroy sharp corners (like the vertices of 'v', 'w', and crossbars of 't').

- Fits local 3rd-order polynomials over sliding 7-point windows.
- Suppresses coordinate quantization noise without rounding out sharp stroke cusps.

4. Attention-Verified Sampling

0 Dropped Glyphs

Recurrent attention occasionally stalls on repetitive double letters ('ll', 'ee') or skips trailing punctuation.

- **Automated Verification:** Enforces minimum character soft-window dwell count ($\min_u \text{count}(u) \geq 6$).
- **Aspect Ratio Guard:** Rejects degenerated tight-loop attractors (requires width/height > 2.0).

Verified Synthesis Outputs & Benchmarks

Multi-line notebook page composition, unit test coverage, and live generated sample.

Live Model Vector Output (Real SVG)

Ruled Notebook Paper

Direct vector rendering of `Online Handwriting Synthesis` & `GLA University Capstone 2026` (1,744 stroke points, C^1 Bézier curves):



Line Height: 46.0 units | Red Margin Rule: #e8a0a0 | Ink Color: #0f172a

System Health & Test Dashboard

Passing 100%

70 / 70	-2.155	< 3.2s
PYTEST SUITE	VALIDATION NLL	FULL PAGE GEN

Complete automated coverage across geometry transformations, XML parsers, ragged HDF5 loaders, attention alignment invariants, and Bézier SVG path generation.

Composable Document Synthesizer

`scripts/synthesize.py`

- **Automated Word-Wrapping:** Splits arbitrary length text across lines while preserving full word boundaries.
- **Indian School Notebook Rulings:** 42–48 unit horizontal guides with vertical red margin rule (`stroke="#e8a0a0"`).
- **Baseline Calibration:** Anchors letters using 20th-percentile bounding box, ensuring natural descenders ('g', 'p', 'y').

Project Roadmap & Next Milestones

Clear demarcation of completed work, active development, and upcoming deliverables.

Completed Milestones (100% Done)

Phases 0–4

- **Dataset Preprocessing:** Ingestion of 12,178 IAM-OnDB lines, spatial resampling, and ragged 1D HDF5 dataset pipeline.
- **Dual-Mode Tokenizer:** Linear move bins + asinh jump bins for numerically stable Cross-Entropy sequence training.
- **Canonical Graves 2013 RNN:** Full 3-layer peephole LSTM with soft-window Gaussian attention reproduced (-2.155 NLL).
- **Transformer Decoder:** 20 full epochs trained with causal self-attention and writer style conditioning.
- **Calligraphic Spline Engine:** Catmull-Rom cubic Bézier renderer with temperature-scaled sampling (T=0.35).
- **Official Academic SRS:** 10-page GLA SRS specification finalized, signed off, and approved.

In Progress & What's Left

Next Sprints

- **Sprint 5 — Stylus / Tablet Input Canvas:** Real-time browser canvas to capture user stylus stroke vectors for few-shot handwriting style cloning.
- **Sprint 6 — Web UI Playground:** Interactive web app (React / Vite + FastAPI backend) allowing live document synthesis, custom ink styling, and PDF export.
- **Sprint 7 — Quantitative Recognition Ablation:** Word Error Rate (WER) and Character Error Rate (CER) benchmarking using pretrained offline TrOCR.
- **Sprint 8 — Final Thesis & Viva Defense:** Final comprehensive capstone documentation, presentation, and oral defense.

Core Vector Generative Engine Operational • Zero Regressions • Prepared for Faculty Review & Live Demonstration

GLA University • Dept of CEA (AI)

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